Efe **Yazgan**

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Current and Past Roles

Geneva, Switzerland

Coordinator/Convener/PI Roles

2007 - now

- LHC Physics Center Monte Carlo Working Group CMS *Convener* — 2026 – present
- *Co-convener (L3)* of the CMS Top Quark + X + Bosons Physics Group — 2025 – present
- *Coordinator* of the NTU CMS analysis group for extra Yukawa couplings and new Higgs boson searches — 2020 – present
- *Co-PI* Decadal Mission for the New Higgs/Flavor Physics Era of NSTC Grant 114-2639-M-002-007 — 2025 – 2027
- *Co-PI* Decadal Mission for the New Higgs/Flavor Physics Era of NSTC Grant 113-2639-M-002-006 — 2024 – 2025
- *PI* Search for new Yukawa couplings at the LHC with heavy quarks of NSTC Grant 1110-2112-M-002-017 — 2021 – 2023
- *Co-convener* of the High Energy Physics Software Foundation (HSF) Physics Generators group — 2020 – 2023
- *Co-convener (L2)* of the CMS Monte Carlo Event Generators — 2017 – 2019
- *Co-convener (L3)* of the CMS Physics Comparisons and Generator Tunes — 2016 – 2017
- *Co-convener (L3)* of the CMS Top Quark Event Modelling Generators — 2015 – 2017
- *Co-convener (L3)* of the CMS Top Quark Properties — 2013 – 2014
- *Co-convener (L3)* of the CMS Electroweak Dilepton physics group — 2010 – 2011
- *Co-coordinator* of the CMS Hadron Calorimeter (HCAL) Data Certification detector group — 2009 – 2011
- *Co-coordinator* of the HCAL Prompt Analysis detector group — 2008 – 2009

Geneva, Switzerland

Reviewer, Editor, Committee, Consultant Tasks

2007 - now

- *Reviewer* for the journal *IEEE Transactions on Nuclear Science* — 2026
- *Review Committee Chair (for a CMS generator physics analysis)* — 2026 – present
- *Review Committee Chair (for a CMS top quark analysis)* — 2024 – 2026
- *Review Committee member* for > 20 CMS Higgs, top, and electroweak, and modeling studies — 2012 – present
- *Editorial Board Member* of the Physics Section of the *Symmetry (MDPI)* journal — 2024 – present
- *Reviewer* for the journals *Particles, Galaxies, and Symmetry (MDPI)* — 2025 – present
- Assessment of self-citations of manuscripts for the *Symmetry* journal — 2025
- *Co-editor* of the special journal issue *New Higgs-like states* in the *Symmetry* journal — 2024 – present
- *Co-editor* of the special journal issue *Top quark at the new physics frontier* in the *Universe (MDPI)* journal — 2021 – 2024
- *Reviewer* for the journal *Physics of the Dark Universe (ELSEVIER)* — 2023
- *Reviewer* for *QEIOS* for an astrophysics preprint — 2023
- *Book reviewer* for *CRC press (Routledge)* — 2022
- *Reviewer* for the *European Physical Journal C* — 2019 – 2020
- *Reviewer* for *Physics Letters B (ELSEVIER)* — 2017 – 2018, 2020
- *Jury member* for the Ph.D. thesis of Deniz Poyraz (Ghent University) — 2018
- *Consultant* for the project *Free electron distribution in the Galaxy* (TUBITAK) — 2015 – 2018
- *Jury* and the *reading committee* for the Ph.D. thesis of Kelly Beernaert (Ghent University) — 2015
- *Scientific advisor* for the Turkish translation of the book *Albert Einstein and Relativity for Kids* (TUBITAK) — 2015
- *Operations research committee* for *Izmir HEP Workshop; New Physics at the LHC and Turkey-CERN relations* — 2013
- *Co-editor* of the Book *Neutron Stars, Supernovae and Supernova Remnants*, *Nova* — 2007

Geneva, Switzerland

Contact person

2007 - now

- Harmonization of Modeling Uncertainties in ATLAS & CMS Top Quark Measurements in the TOPLHC WG — 2021 – present
- CMS Event Simulation Sample Verification Script contact — 2019 – present
- HEP Software foundation Physics Generators Working Group CMS Contact person — 2019 – 2023
- CMS Top quark conference contact person — 2014 – 2019
- CMS HCAL contact for global runs, particle flow, and the physics validation team — 2007 – 2011
- CMS Calorimeter noise contact person in V+Jets group — 2010
- CMS Physics Validation team contact person for HCAL — 2009 – 2011
- Data reconstruction/transfer for CMS ECAL/HCAL Test beam2006 to set up the 1st remote ops. center of CMS — 2007 – 2008

Conference/Workshop Tasks

2007 - now

- *Co-organizer*, the HSF Event Generator Tuning Workshop, Virtual — Jun. 2023
- *Co-organizer*, the MC4EIC 2022, workshop on Monte Carlo event simulation for the EIC, Virtual — Nov. 2022
- *Chair*, the parallel session "Standard Model Physics at the TeV Scale" in PANIC2021 Conference, Virtual — Sep. 2021
- *Co-organizer*, the HEP Software Foundation Worldwice LHC Computing Grid (WLGC), Virtual — Nov. 2020
- *Local organizing committee*, 12th International Workshop on Top Quark Physics, Beijing, China — Sep. 2019
- *Chair*, Plenary: FCNC and EFT, 11th International Workshop on Top Quark Physics, Bad Neuenahr, Germany — Sep. 2018
- *Organizing Committee/Simulation Session Convener*, CMS Heavy flavour tagging workshop, Brussels, Belgium — Apr. 2018
- *Theory and Modeling Session Convener* in TOP@100 fb^{-1} Workshop, CERN, Geneva, Switzerland — Nov. 2016

Selected publications (in the last five years)

Journal papers

- Search for charged Higgs bosons decaying into top and bottom quarks in lepton+jets final states in proton-proton collisions at $\sqrt{s}$ = 13 TeV, *Jour. High Ene. Phys.* 10 (2026) 013; arXiv:2512.24471
- Review of top quark mass measurements in CMS, *Phys. Rep.* 1115 (2025) 116
- Top Quark at the New Physics Frontier, Eds. E. Yazgan and P. Silva., (2024) doi.org/10.3390/books978-3-7258-0614-0
- Search for new Higgs bosons via same-sign top quark pair production in association with a jet in proton-proton collisions at $\sqrt{s}$ = 13 TeV, CMS Collaboration, *Phys. Lett. B.* 850 (2024) 138478
- CMS PYTHIA 8 colour reconnection tunes based on underlying-event data, CMS Collaboration, *Eur. Phys. J. C* 83 (2023) 587
- Distances of galactic radio pulsars; first quadrant: $-2 < \ell < 90$ and $-2 < b < 2$, P. Kutukcu, A. Ankay, E. Yazgan, and K. Bozkurt, *Mon. Not. Roy. Astro. Soc.* 511 (2022) 4669
- Challenges in Monte Carlo Event Generator Software for High-Luminosity LHC, A. Valassi, E. Yazgan, J. McFayden et al., *Comput. Softw. Big Sci.* 5 (2021) 12

Preprints and Proceedings

- Measurements of top quark properties in CMS: $t\bar{t}$ spin density matrix, quantum entanglement and quantum magic, E. Yazgan, arXiv:2510.13743 (2025)
- Search for charged Higgs bosons decaying into a top and a bottom quark in proton-proton collisions at $\sqrt{s}$ = 13 TeV, CMS Collaboration, CMS-PAS-B2G-24-008 (2025)
- Input to European Strategy Update for Particle Physics: Sustainability, V. Boisvert et al. arXiv:2504.03012 (2025)
- Event Generators for High-Energy Physics Experiments, J. M. Campbell et al. arXiv:2203.11110 (2022 – 2025)
- TF07 Snowmass Report: Theory of Collider Phenomena, F. Maltoni, et al. arXiv:2210.02591, FERMILAB-FN-1203-QIS (2022)
- Modelling the Data at the LHC, E. Yazgan, arXiv:2110.02599, CMS CR-2021/166; PoS (PANIC2021) 410 (2021)
- HL-LHC Computing Review Stage-2, Common Software Projects: Event Generators, E. Yazgan, et al. arXiv:2109.14938 [hep-ph, hep-ex, physics.comp-ph]; FERMILAB-PUB-21-526-OCIO-SCD-T (2021)

Link to full list of selected publications • **Link to full list of (~ 1400) publications** • **Link to preprints in arXiv**

Selected presentations (in the last five years)

Toponium Searches in ATLAS and CMS, — *Hvar, Crotia*
LHC Days in Split/Hvar 2026, (Invited talk) — *21–25 September 2026*

Event engineering techniques for small systems in CMS, — *Torino, Italy*
EMMI Workshop: QCD Challenges from pp to AA collisions, (Invited talk) — *6–10 July 2026*

Charged Higgs boson searches at the LHC, — *Geneva, Switzerland*
Extended Scalar Sectors From All Angles 2026 workshop (Talk on behalf of the CMS Collaboration) — *1–5 June 2026*

Top quark properties in CMS: $t\bar{t}$ spin density matrix, quantum entanglement, and quantum magic — *Marseille, France*
EPS-HEP Conference (Talk on behalf of the CMS Collaboration) — *7 – 11 July 2025*

Searches for Additional Scalar Bosons with the CMS Experiment — *Paris, France*
Higgs Hunting 2024 Workshop (Plenary Talk on behalf of the CMS Collaboration) — *23 – 25 September 2024*

ATLAS and CMS Top Quark Rare Searches (Flavor Changing Neutral Currents) — *Rome, Italy*
Standard Model at the LHC 2024 Workshop (Plenary Talk on behalf of the ATLAS & CMS Collaborations) — *7 – 10 May, 2024*

CMS Highlight talk: Search for extra Higgs bosons through same-sign top-quark production in association with an extra jet — *Fully online workshop*
LHC TOP Working Group Meeting (Plenary Talk on behalf of the CMS Collaboration) — *29 Nov – 1 Dec, 2023*

Modelling the data at the LHC — *Fully online conference*
PANIC2021 (22nd Particles and Nuclei International Conference (Plenary talk on behalf of the ATLAS & CMS Collaborations) — *5 – 10 September 2021*

Hyperlink to full list of presentations

Positions

Research faculty (based at CERN) — *Geneva, Switzerland*
National Taiwan University, Taipei, Taiwan — *2020 - PRESENT*

Post-doc fellow (based at CERN) — *Geneva, Switzerland*
Institute of High Energy Physics (IHEP), Chinese Academy of Sciences, Beijing — *2016 - 2020*

Post-doc fellow (based at CERN) — *Geneva, Switzerland*
University of Ghent, Belgium — *2011 - 2016*

Postdoctoral research associate (based at CERN) — *Geneva, Switzerland*
Texas Tech University, TX, USA — *2007 - 2011*

Guest scientist and software developer — *Batavia, IL, USA*
Fermi National Accelerator Laboratory (FERMILAB) — *2005 - 2007*

Research and teaching assistant — *Ankara, Turkey*
Middle East Technical University (METU) — *2000 - 2007*

Education

Middle East Technical University with host laboratory: Fermilab — *Ankara, Turkey / Batavia, IL, USA*
PhD in Physics — *2002 - 2007*

- GPA: 3.61/4.00, Mentors: Shuichi Kunori and Dan Green; Thesis: "Search for a Standard Model Higgs Boson in CMS via Vector Boson Fusion in the $H \rightarrow W^+W^- \rightarrow \ell^+\ell^-\nu\bar{\nu}$ Channel and Optimization of Energy Reconstruction in CMS using Test Beam 2006 Data" fermilab-thesis-2007-13.

Middle East Technical University — *Ankara, Turkey*
MSc in Physics (Specialisation in Astrophysics) — *2000 - 2002*

- GPA: 3.56/4.00; Supervisor: S. Balman. Thesis: "Timing Characteristics of Intermediate-Polar Cataclysmic Variables and Data Analysis of YY Draconis"

Middle East Technical University — *Ankara, Turkey*
BS in Physics — *1996 - 2000*

- GPA: 3.31/4.00; Undergraduate special advanced physics class

Honors & Awards

2025 — **Breakthrough Prize in Fundamental Physics** , *awarded to E. Yazgan and colleagues in the CMS Collaboration at the LHC* for detailed measurements of Higgs boson properties confirming the symmetry-breaking mechanism of mass generation, the discovery of new strongly interacting particles, the study of rare processes and matter-antimatter asymmetry, and the exploration of nature at the shortest distances and most extreme conditions at CERN's Large Hadron Collider. — *Los Angeles, USA*

2025 — **Exceptional Reviewer 2025**, for the $Symmetry$ Journal — *Basel, Switzerland*

2019 — **CMS achievement award**, for work in the validation tools for generators and for sample production — *Geneva, Switzerland*

2018 — **Outstanding Contribution in Reviewing** , from the Editors of Physics Letters B — *Amsterdam, Netherlands*

2013 — **European Physical Society High Energy and Particle Physics Prize**, *awarded to the CMS Collaboration* for the discovery of a Higgs boson, as predicted by the Brout-Englert-Higgs mechanism — *Stockholm, Sweden*

2007 — **CMS achievement award**, for crucial contributions to analyses of ECAL+HCAL beam tests and global runs — *Geneva, Switzerland*

Supervision

T.-H. Hsu "Extra Yukawa and new Higgs boson searches in Run 2 and Run 3 with g2HDM" (PhD mentor)

Geneva, Switzerland

National Taiwan Univ. 2020 – present

H.-Y. Yeh "Extra Yukawa and new Higgs boson searches in Run 3 with g2HDM" (MSc mentor)

Geneva, Switzerland

National Taiwan Univ. 2025 – present

N. Bulte, "Search for Scalar Top Quark Partners & Parton Shower Tuning in Modelling Top Quark Pairs" (MSc supervisor)

Geneva, Switzerland

Ghent Univ. 2016

K. Beernaert, "Spin Correlations in $t\bar{t}$ Systems in the μ+jets channel using a Matrix Element Method with the CMS Detector at the LHC" (PhD supervisor)

Geneva, Switzerland

Ghent Univ. 2012 – 2015

B. Bilin, "Measurements of standard model heavy particle production in association with jets using proton-proton collision data at 8 and 13 TeV with the CMS experiment at the LHC" (PhD mentor)

Geneva, Switzerland

Middle East Tech. U. 2011 – 2016

M. O. Sahin, "Search for Z' via Forward Backward Asymmetry in Dielectron Channel with the CMS Detector at the Large Hadron Collider" (MSc mentor)

Geneva, Switzerland

Middle East Tech. U. 2011 – 2012

B. Bilin, "Production of Z+Jet Events and HF Jet Energy Scale Calibration at 7 TeV in the CMS Experiment at LHC" (MSc mentor)

Geneva, Switzerland

Middle East Tech. U. 2010 – 2012

Y. J. Roh, "Measurement of the Forward-Backward Asymmetry in ɣ/Z Boson to Dilepton Events in Compact Muon Solenoid at a Center-of-mass Energy of 7 TeV" (PhD mentor)

Geneva, Switzerland

Texas Tech. U. 2009 – 2011

Geneva, Switzerland

Supervised summer projects at CERN

- B. Ors (Ecole Jeannine Manuel, Paris), *Job Shadowing* in CMS 2025
- G. Vandierendonck (Ghent Univ.) "Four Top Quark Production at the High Luminosity LHC" 2016
- N. Bulte (Ghent Univ.), "Determination of the Top Quark Pole Mass at √s= 7,8, and 13 TeV" 2015
- I. Michiels (Ghent Univ.) , "$t\bar{t}$ Spin Correlation in the $t\bar{t}H$ Channel" 2014
- S. Demulder (Vrije Univ. Brussels), "Can the Incline Asymmetry be Observed at the LHC" 2014
- J. Duvillier (Ghent Univ.), "Energy Asymmetry Measurement using $t\bar{t}$+jet Systems at $\sqrt{s}$=8 TeV in CMS" 2014
- C. Moortgat (Ghent Univ.), "Effects of Non-Zero Chromo-Moments on $t\bar{t}$ Spin Correlation 2013
- M. Unal (CERN, Geneva), "Analyses of Z+1 Jet Events using the Full Rapidity Coverage of CMS" 2012
- M. Orlov (CERN, Geneva), "Validation of Single Particle Response in HCAL" 2008

Geneva, Switzerland

Supervised postdocs

- Gouranga Kole, "Extra Yukawa and new Higgs boson searches with g2HDM" 2022 – present
- Agostino De Iorio, "Extra Yukawa and new Higgs boson searches with g2HDM" 2023 – present

Geneva, Switzerland

Coach

- high school students for the CERN/DESY beamline for schools competition 2021

Teaching

Lectures at Biricik Bilim High Energy Physics School — *Online*

Top quark physics — *2026 – 2026*

Lectures at CERN Teachers Programs (TTP) — *Geneva, Switzerland*

Astroparticle physics (Neutrinos, astroparticle physics, gravitation and cosmology) — *2015 – 2020*

Lectures at the Hadron Collider School (HASCO) — *Gottingen, Germany*

Top quark physics — *2014 – 2017*

Lectures at ERASMUS ISSCSMB-12, International Summer School and Conference on High Energy Physics: Standard Model and Beyond — *Belek/Antalya, Turkey*

Top quark physics at colliders — *28 Aug – 8 Sep 2012*

Problem sessions and laboratory courses at METU — *Ankara, Turkey*

Graduate electromagnetic theory I, physics for students in Science I & II, concepts of modern physics, freshmen physics laboratory I & II, experiments in physics — *2000 – 2005*

Tutorials

Tutorials on the two main tools in Monte Carlo production: Monte Carlo request Management and Production Monitoring Platform — *Geneva, Switzerland* — *2017 – 2018*

Data Quality Monitoring (DQM) shift tutorials — *Geneva, Switzerland* — *2009 – 2012*

Science Communication

- Translation of **Particle Physics for Babies** (by Louie D. Corpe) to Turkish with Y. Uzunefe Yazgan., 2025
- **The best non-technical books to get interested in knowing the Universe** published in *Shepherd.com*, 2024
- **Towards the New Physics Higgs/Flavor Era**, poster in *ASP March meeting*, Taiwan. 2024
- **Hunting the Higgs Boson Siblings with Top Quarks**, *CMS Physics Briefing*, 2023
- **Higgs physics**, presentation in *Turkey-CERN status review*, CERN, Geneva, Switzerland, 2022
- **Search for extra Higgs Bosons at the LHC**, E. Yazgan et al. poster in *ASP March meeting*, Taiwan. 2022
- **Data science in high energy particle physics**, invited talk in *Istanbul Tech Week* (online conference), 2020
- **Old School Electroweak** in *CMS Result of the Month in Fermilab Today*, 2012
- Poster for **CERN: Accelerating Science Exhibition**, METU, Ankara, Turkey. 2012
- **Interview about CERN**, published in the journal *Radikal*, and *Matematik Dünyasi*, 86 (2011) 20. 2011
- **Rediscovering the Weak Force** in *CMS Result of the Month in Fermilab Today*, 2010
- **Interviews on the *LHC First Beam Day*** on *CNN Turk*, 2008
- **Interview about CERN**, **Experiment of the Century**, published in the journal *Radikal*, 2007
- **Nobel physics prizes in astrophysics**, *Bilim ve Utopya*, 2003
- *METU Physics Department* **Outreach Days for High School Students**; Ankara, 2002, 2003

Programming and software

- Python, C/C++, Fortran, Shell, R, Bash, ...
- ROOT, CMSSW, Git, JIRA, Event generators, Mathematica, ...
- Latex, HTML/CSS, Markdown, ...

Languages

English, French, Turkish